

TABLE 3
LIDAR DATASHEET

Item	Min	Typical	Max	Unit	Remarks
Ranging frequency	/	3000	/	Hz	Ranging 3000 times per second
Motor frequency	5	6	8	Hz	Need to connect to PWM signal, recommended to use the speed of 6Hz
Ranging distance	0.12	/	8	m	Indoor environment with 80% reflectivity
Field of view	/	0-360	/	Deg	/
Systematic error	/	2	/	cm	Range ≤ 1m
Relative error	/	3.5%	/	/	1m < Range ≤ 6m
Tilt angle	0.25	1	1.75	Deg	/
Angle resolution	0.60 (frequency @5Hz)	0.72 (frequency @6Hz)	0.96 (frequency @8Hz)	Deg	Different motor frequency

Note: (1) It is factory FQC standard value, 80% reflectivity material object. (2) The relative error value indicates the accuracy of the lidar measurement.

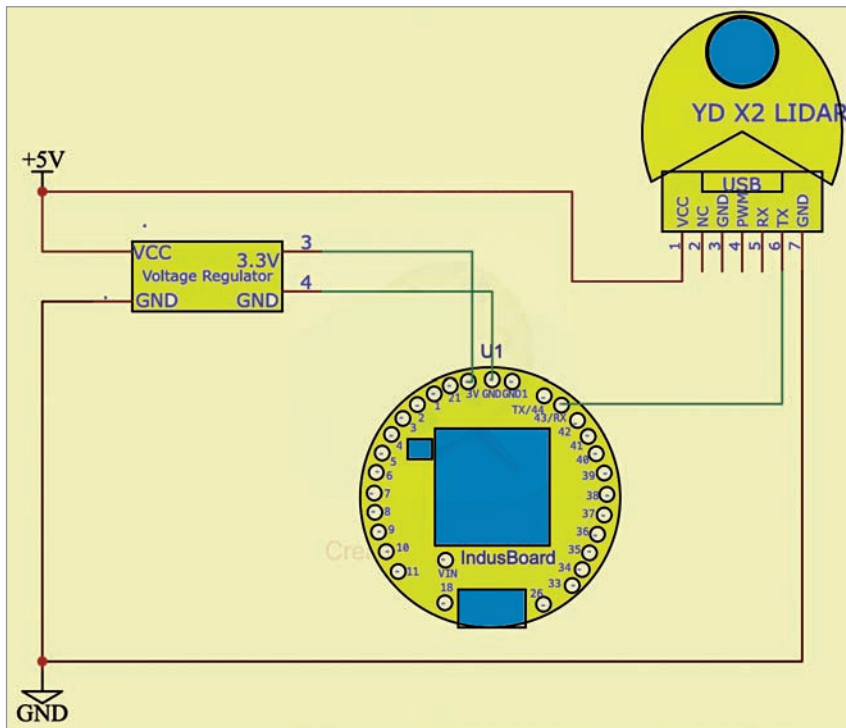


Fig. 4: Circuit diagram

ing, and similar tasks. We combined the IndusBoard with a lidar unit to scan surroundings, generate data, and create real-time wireless maps displayed on a web page. Fig. 1 shows the IndusBoard fixed to the lidar. The components for the project are listed in Table 1. Fig. 2 shows

the lidar used during testing.

You can select any lidar model; in this project, we used the YD X2 lidar. According to the datasheet, it can scan 360 degrees and measure up to 8 metres, but other lidars can measure up to 1km. Choose based on your requirements.

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